Shared resources

Root defined a guild as a group of species that “exploit the same class of environmental resources in a similar way.” It does not matter whether species in a guild are related or not—all that matters is how they use their environment. They do not even have to occupy the same niche; they just have to use the same resource.

Guilds are typically identified by the food resource they have in common, although it could be any other resource that they share. Sharing the same resource means that guild members often compete with one another, but they are not necessarily in constant competition. For example, although they may compete for the same food, on other occasions they might cooperate to deal with predators.

The guild concept was a major breakthrough in thinking about connections between organisms in ecosystems. The theory implied that the entire functioning of an ecosystem could be understood by identifying all the guilds within it. Although that was potentially a huge undertaking, ecologists have now managed to identify many more guilds that confirm links between species. For example, the birds of North America can be grouped into guilds of gleaners, excavators, hawkers, aerial chasers, and scavengers.



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